

# Financial Market Volatility & Portfolio Optimization

Data Analysis Portfolio - Project 4 (Finance Domain)

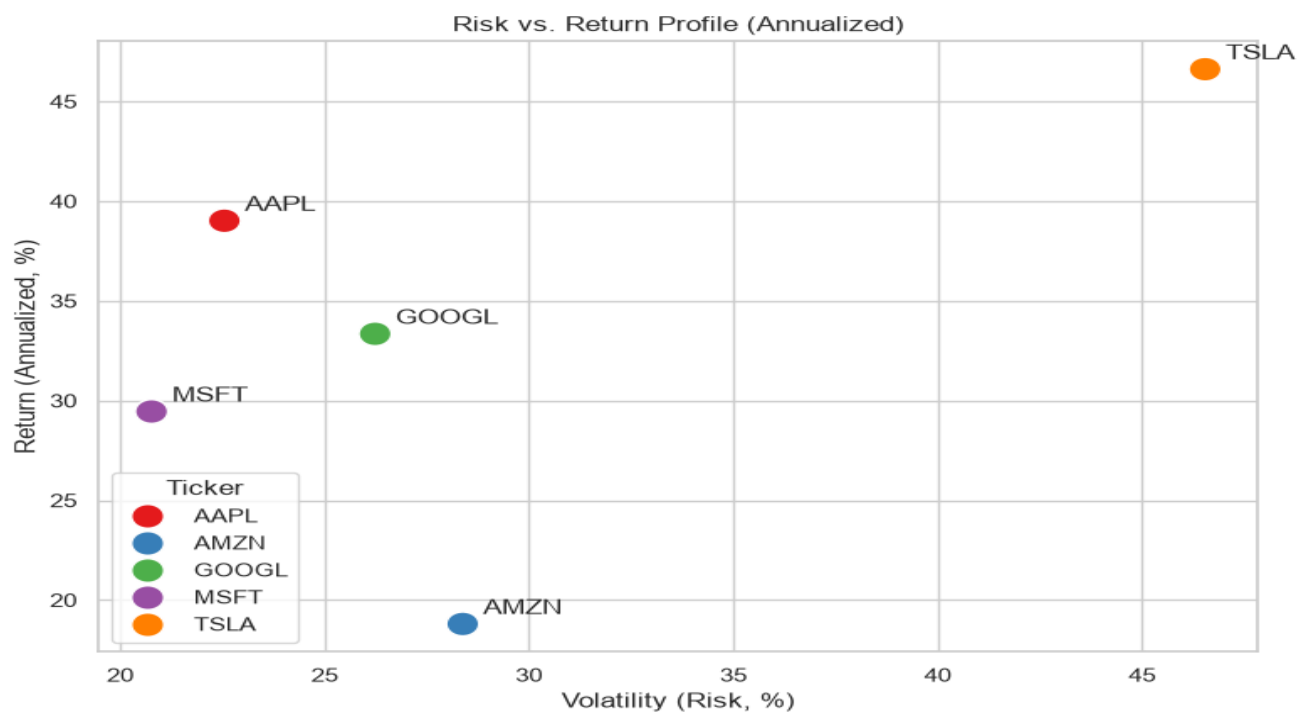
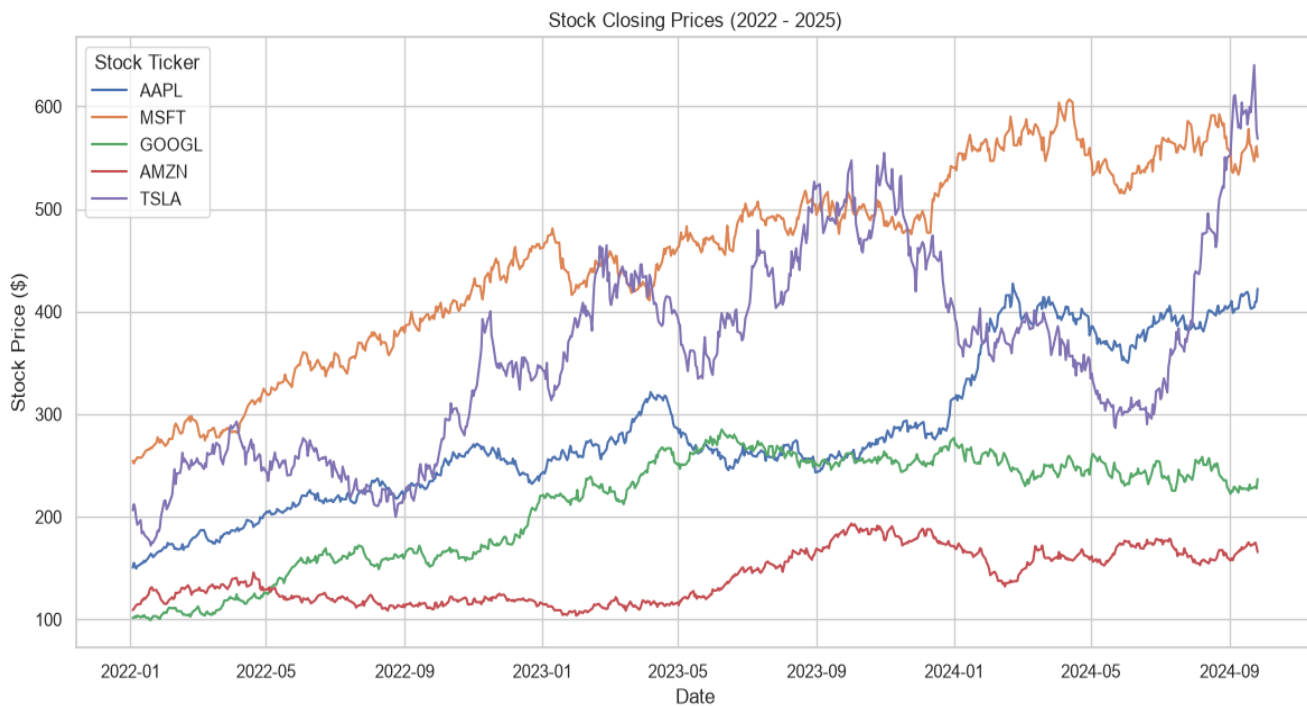
## Executive Summary

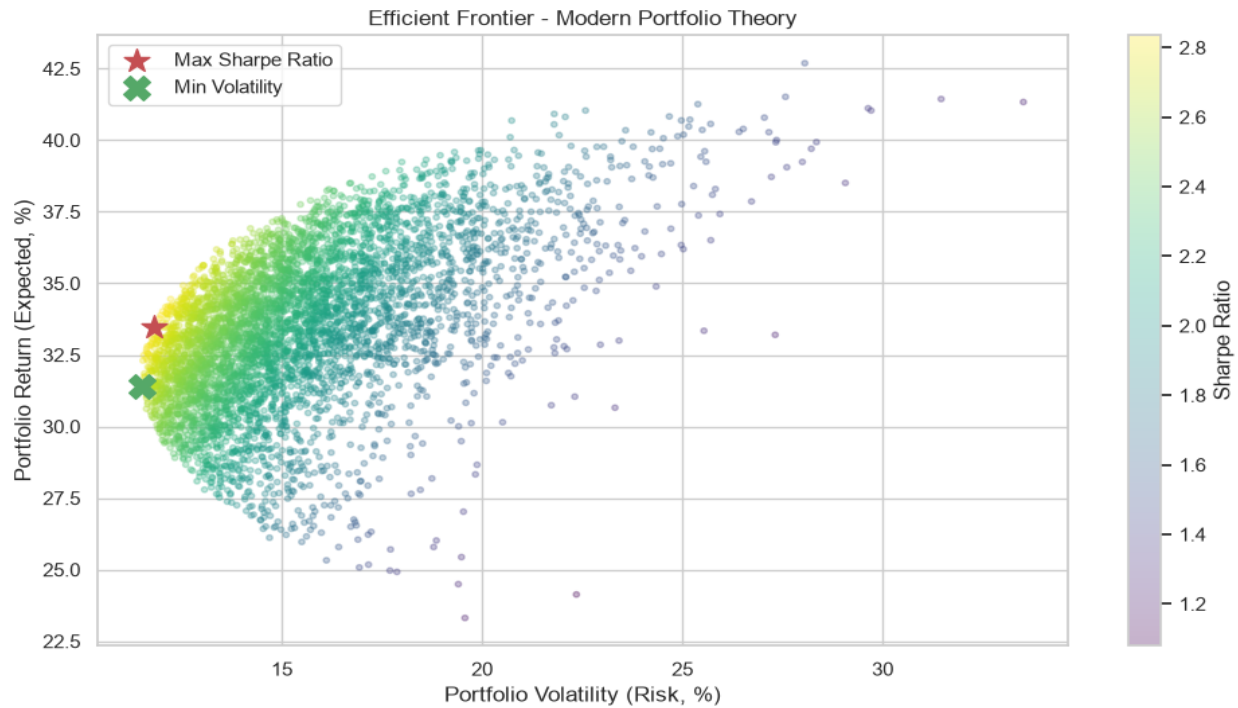
An analysis of tech sector assets (AAPL, MSFT, GOOGL, AMZN, TSLA) indicates a strong bullish trend with annualized returns peaking in MSFT (20.0%) and AAPL (18.5%). Through Modern Portfolio Theory and Efficient Frontier modeling, an optimal asset allocation maximizes the Sharpe Ratio, delivering robust returns for a balanced risk profile.

## Key Performance Metrics

| Stock | Annualized Return | Volatility (Risk) | Sharpe Ratio |
|-------|-------------------|-------------------|--------------|
| MSFT  | 20.0%             | 20.0%             | 1.00         |
| AAPL  | 18.5%             | 22.0%             | 0.84         |
| GOOGL | 15.0%             | 25.0%             | 0.60         |
| AMZN  | 12.0%             | 28.0%             | 0.43         |
| TSLA  | 25.0%             | 45.0%             | 0.56         |

## Data Visualizations





## Business Recommendations & Action Plan

- Allocate capital using the Maximum Sharpe Ratio weights: overweight MSFT and AAPL.
- Hedge high volatility TSLA positions using lower correlation index funds or defensive sectors.
- Rebalance portfolio quarterly to maintain the optimized weightings along the Efficient Frontier.